

6 Conclusion

This paper provides causal evidence that the adoption of artificial intelligence significantly enhances productivity in European firms, while simultaneously reshaping our understanding of its labor market implications. The results show that the adoption of AI increases labor productivity by 4% on average – a substantial yet measured gain that aligns with mid-range macroeconomic projections rather than more transformative estimates. Crucially, the productivity improvements stem from capital deepening rather than workforce reduction: AI augments worker output without diminishing employment levels in the short term, countering prevalent concerns about immediate job displacement. Moreover, the evidence suggests that, so far, workers in AI-adopting firms have benefited in terms of higher wages. The productivity benefits are, however, disproportionately concentrated among larger firms, financially advanced regions and technology-intensive sectors, risking heightened economic polarization.

The findings yield two key policy directions. First, targeted financial support mechanisms can help companies grow and reach the critical mass to benefit from AI transformation. Our analysis shows that firms in an environment with more sophisticated financial market options are better equipped to invest in AI. Therefore, developing more effective and sophisticated financial markets is a priority. This underscores the importance of focusing on the EU Savings and Investment Union, which is crucial for fast-growing, innovative companies. Second, workforce development can prioritize human-AI complementarity through retraining and upskilling programs that upgrade worker skills towards prompt engineering, data stewardship and AI-augmented decision-making. This requires coordinated investments in vocational training, tertiary education and lifelong learning programs that will equip workers with the technical and cognitive capabilities needed to collaborate effectively with AI systems.

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